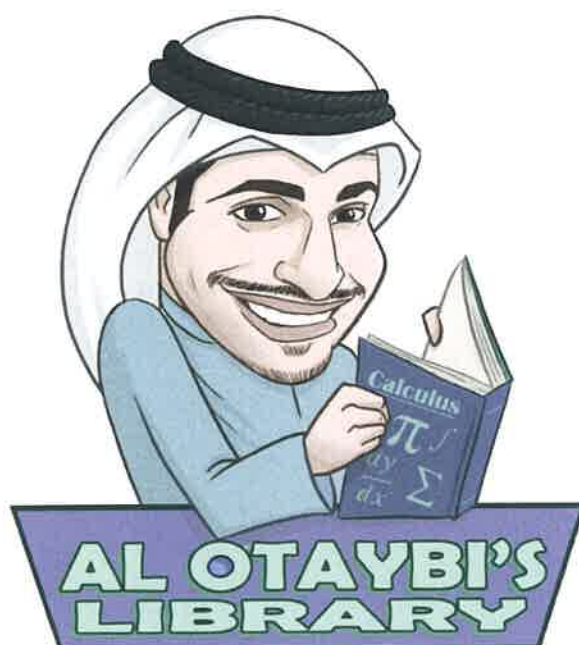


Kuwait University

2021/2022

Fall course



Calculus A

Second

Sections : 3.9

Related Rates

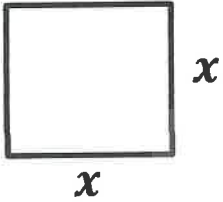
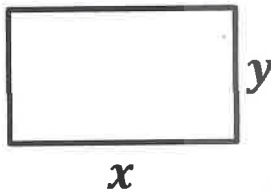
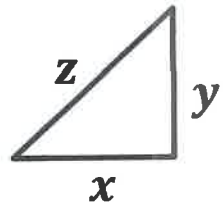
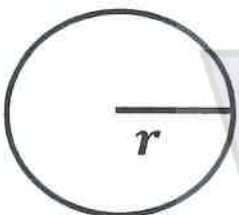
(12)

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi



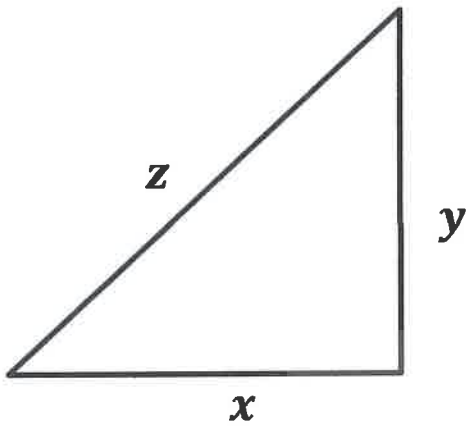
3.9 Related rates



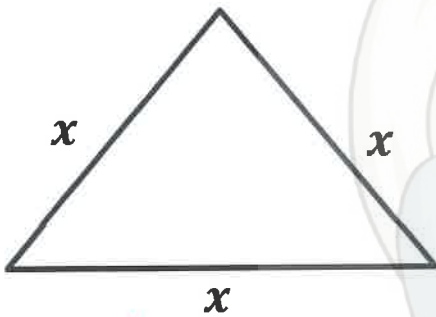
		Area المساحة	Perimeter or (Circumference) المحيط
	Square مربع	$A = x^2$	$C = 4x$
	Rectangle مستطيل	$A = x \cdot y$	$C = 2x + 2y$
	Right triangle مثلث قائم الزاوية	$A = \frac{1}{2} x \cdot y$	$C = x + y + z$
	Circle دائرة	$A = \pi \cdot r^2$	$A = 2 \cdot \pi \cdot r$



Special laws for Triangle



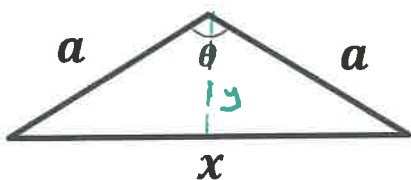
$$z^2 = x^2 + y^2$$



equilateral triangle
مثلث متساوي الأضلاع

$$A = \frac{\sqrt{3}}{4} x^2$$

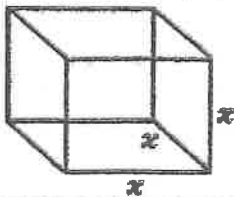
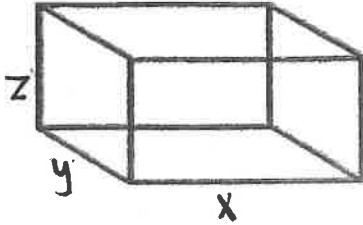
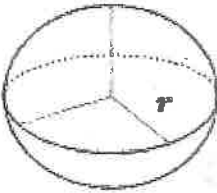
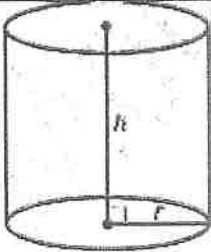
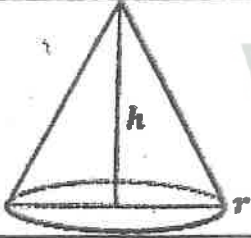
$$C = 3x$$



Isosceles triangle
مثلث متساوي الضلعين

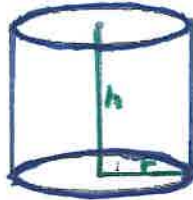
$$A = \frac{1}{2} xy \quad \text{or} \quad A = \frac{1}{2} a^2 \sin \theta$$

$$C = 2a + x$$

		Volume (الحجم)	Surface Area (مساحة السطح)
	Cube (مكعب)	$V = x^3$	$S = 6x^2$
	Rectangular Box	$V = x \cdot y \cdot z$	$S = 2x \cdot y + 2x \cdot z + 2y \cdot z$
	Sphere (كرة) (ball)	$V = \frac{4}{3} \cdot \pi \cdot r^3$	$S = 4\pi \cdot r^2$
	Right Circular Cylinder (اسطوانة دائرية قائمة)	$V = \pi \cdot r^2 \cdot h$	$S = 2\pi \cdot r \cdot h$
	Cone (مخروط)	$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$	$S = \pi \cdot r \cdot \sqrt{r^2 + h^2}$

(اسطوانة دائرية قائمة) Right Circular Cylinder

Open top and bottom



$$S = 2\pi \cdot r \cdot h$$

$$V = \pi \cdot r^2 \cdot h$$

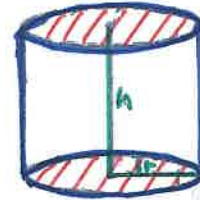
Open top (closed bottom)



$$S = 2\pi \cdot r \cdot h + \pi \cdot r^2$$

$$V = \pi \cdot r^2 \cdot h$$

Closed top and bottom



$$S = 2\pi \cdot r \cdot h + 2\pi \cdot r^2$$

$$V = \pi \cdot r^2 \cdot h$$

(مخروط) cone

Open bottom



$$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$$

$$S = \pi \cdot r \cdot \sqrt{r^2 + h^2}$$

Closed bottom

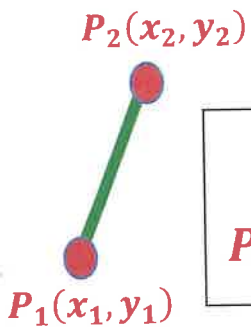


$$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$$

$$S = \pi \cdot r \cdot \sqrt{r^2 + h^2} + \pi \cdot r^2$$



قوانين لا علاقة لها بالأشكال



Distance between point

$P_1(x_1, y_1)$ and $P_2(x_2, y_2)$

$$l = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$l^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$



Distance by time

الزمن . السرعة = المسافة
(rate)

$$x = \frac{dx}{dt} \cdot t$$



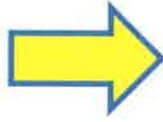
سوالف !!



الوحدات

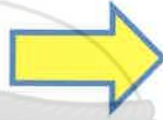


Volume



(طول)³
 $cm^3 / m^3 / ft^3$

Area / Surface area



(طول)²
 $cm^2 / m^2 / ft^2$

Length
(altitude-side-radius..)



(طول)
 $cm / m / ft$



Key word



Rate

Speed

Velocity

fast



$$\frac{d \square}{dt}$$



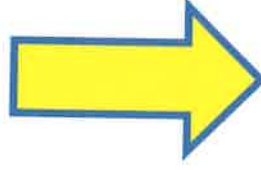


Warning !!



increasing

أو أي كلمة تدل على الزيادة
(heating.....)

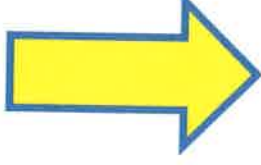


يكون الـ **rate**

موجب +

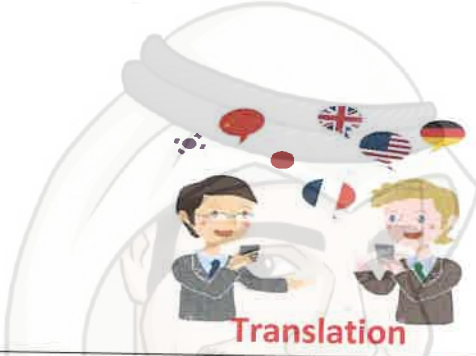
decreasing

أو أي كلمة تدل على النقصان
(melting.....)



يكون الـ **rate**

سالب -



الكلمة	الترجمة
Length	طول
Altitude – Height	ارتفاع
Base	قاعدة
Side – edge	ضلع
Radius	نصف قطر
Diameter	قطر
Width	عرض
Twice	ضعف
intersection	تقاطع

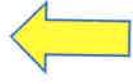


خطوات الحل



(1) الرسم (ان امكن).

$$\frac{d \boxed{}}{dt}$$



وأعط تحتها خط.

rate

(2) أدور كلمة

(3) كتابة المعطيات والمطلوب.

(4) الربط بين المعطيات والمطلوب باستخدام القانون المناسب.

(5) الاشتقاق بالنسبة للزمن.



A plate in a shape of square is heated . If the area A of the plate (in cm^2) after time t (in hours) is given by $A = \sqrt{1 + t^3}$

Find the rate at which the sides of the plate are changing after two hours.

$$A = \sqrt{1 + t^3}$$

$$\frac{dx}{dt} \quad ??$$

$$t = 2 \text{ hr}$$

$$A = x^2$$

$$\sqrt{1 + t^3} = x^2$$

Diff. w.r.t. t

$$\frac{3t^2}{2\sqrt{1+t^3}} = 2x \frac{dx}{dt}$$

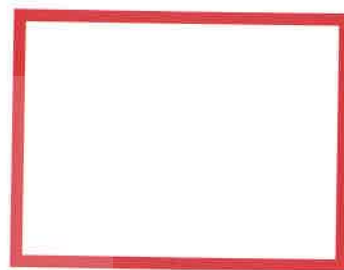
$$\frac{3(2)^2}{2\sqrt{1+8}} = 2\sqrt{3} \frac{dx}{dt}$$

$$\frac{3(4)}{2\sqrt{9}} = 2\sqrt{3} \frac{dx}{dt}$$

$$\frac{3(2)}{3} = 2\sqrt{3} \frac{dx}{dt}$$

$$\div 2\sqrt{3}$$

$$\frac{dx}{dt} = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ cm/hr}$$



x

$$A = x^2 \quad C = 4x$$

$$\text{at } t = 2$$

$$A = \sqrt{1 + 2^3} = \sqrt{9}$$

$$A = 3$$

$$A = x^2$$

$$3 = x^2$$

$$\therefore x = \sqrt{3} \text{ cm}$$

A plate in a shape of disk is heated . If the area A of the plate (in cm^2) after time t (in hours) is given by $A = \sqrt{t^2 + 3t + 6}$ Find the rate at which the radius of the plate is changing after two hours.

$$A = \sqrt{t^2 + 3t + 6}$$

$$\frac{dr}{dt} \quad ?? \longrightarrow t = 2 \text{ hr}$$

$$A = \pi r^2$$

$$\sqrt{t^2 + 3t + 6} = \pi r^2$$

Diff. w.r. t. t

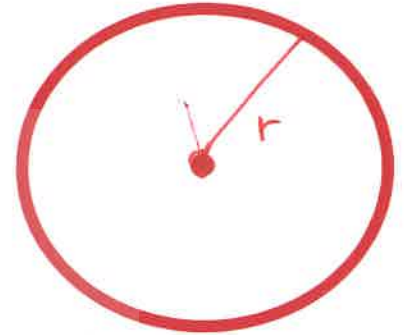
$$\frac{2t + 3}{2\sqrt{t^2 + 3t + 6}} = 2\pi r \frac{dr}{dt}$$

$$\frac{4 + 3}{2\sqrt{4 + 6 + 6}} = 2\pi \frac{2}{\sqrt{\pi}} \frac{dr}{dt}$$

$$\frac{7}{2\sqrt{16}} = 2\sqrt{\pi} \frac{2}{\sqrt{\pi}} \frac{dr}{dt}$$

$$\frac{7}{2(4)} = 4\sqrt{\pi} \frac{dr}{dt}$$

$$\frac{7}{8(4\sqrt{\pi})} = \frac{7}{32\sqrt{\pi}} \text{ cm/hr}$$



$$A = \pi r^2 \quad C = 2\pi r$$

at $t = 2$

$$A = \sqrt{4 + 6 + 6} = \sqrt{16}$$

$$A = 4 \text{ cm}^2$$

$$A = \pi r^2$$

$$4 = \pi r^2$$

$$r^2 = \frac{4}{\pi}$$

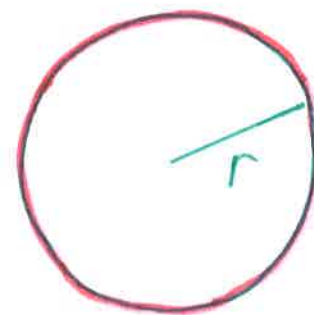
$$r = \sqrt{\frac{4}{\pi}} \rightarrow r = \frac{2}{\sqrt{\pi}}$$

The area of circular region is increasing at a rate of $6\text{cm}^2/\text{sec}$

Find the rate at which its radius changes while its area is 9cm^2 ?

$$\frac{dA}{dt} = 6 \text{ cm}^2/\text{sec}$$

$$\frac{dr}{dt} ?? \longrightarrow A = 9 \text{ cm}^2$$



$$A = \pi r^2 \quad C = 2\pi r$$

$$A = \pi r^2$$

Diff. w.r.t. t

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$6 = 2\pi \left(\frac{3}{\sqrt{\pi}}\right) \frac{dr}{dt}$$

$$6 = 2\sqrt{\pi}\sqrt{\pi} \left(\frac{3}{\sqrt{\pi}}\right) \frac{dr}{dt}$$

$$6 = 6\sqrt{\pi} \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = \frac{6}{6\sqrt{\pi}}$$

$$\frac{dr}{dt} = \frac{1}{\sqrt{\pi}}$$

cm / sec

$$A = \pi r^2$$

$$9 = \pi r^2$$

$$r^2 = \frac{9}{\pi}$$

$$r = \frac{3}{\sqrt{\pi}}$$

A spherical ball made in a steel is heated. If the surface area A of the ball (in cm^2) after time t (in hours) is given by

$$A = \sqrt{t^2 + t - 4}$$

Find the rate at which the radius of the ball is changing after four hours.

$$A = \sqrt{t^2 + t - 4}$$

$$\frac{dr}{dt} ?? \longrightarrow t = 4 \text{ hr}$$

$$A = 4\pi r^2$$

$$\sqrt{t^2 + t - 4} = 4\pi r^2$$

Diff. w.r.t. t

$$\frac{2t+1}{2\sqrt{t^2+t-4}} = 8\pi r \frac{dr}{dt}$$

at $t = 4$

$$A = \sqrt{16 + 4 - 4} = \sqrt{16}$$

$$A = 4 \text{ cm}^2$$

$$A = 4\pi r^2$$

$$4 = 4\pi r^2$$

$$\div 4\pi$$

$$r^2 = \frac{4}{4\pi}$$

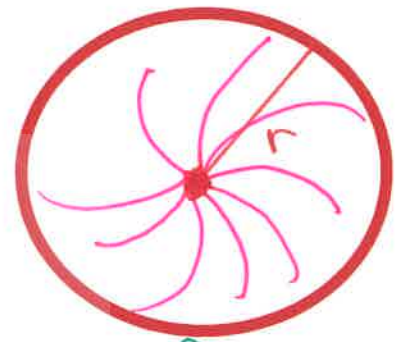
$$r = \frac{1}{\sqrt{\pi}} \text{ cm}$$

$$\frac{8+1}{2\sqrt{16+4-4}} = 8\pi \frac{1}{\sqrt{\pi}} \frac{dr}{dt}$$

$$8 \leftarrow \frac{9}{2(4)} = 8\sqrt{\pi} \sqrt{\pi} \frac{1}{\sqrt{\pi}} \frac{dr}{dt}$$

$$\div 8\sqrt{\pi}$$

$$\frac{dr}{dt} = \frac{9}{8(8\sqrt{\pi})} = \frac{9}{64\sqrt{\pi}} \text{ cm/hr}$$



$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

The altitude of a right triangle is increasing at a rate 1 cm/min while its area is increasing at a rate of $2 \text{ cm}^2/\text{min}$. At what rate is the base of triangle changing when the altitude is 10 cm and the area is 100 cm^2 ?

Dec 2017

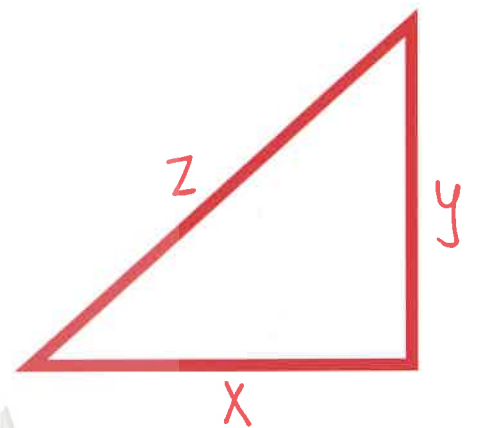
$$\frac{dy}{dt} = 1 \text{ cm/min}$$

$$\frac{dA}{dt} = 2 \text{ cm}^2/\text{min}$$

$$\frac{dx}{dt} ?? \longrightarrow$$

$$y = 10 \text{ cm}$$

$$A = 100 \text{ cm}^2$$



$$A = \frac{1}{2}xy$$

$$C = x + y + z$$

استقاً بـ y + استقاً بـ x

$$A = \frac{1}{2}xy$$

Diff. w.r.t. t

$$(1) \frac{dA}{dt} = \frac{1}{2} (1) \frac{dx}{dt} y + \frac{1}{2} x (1) \frac{dy}{dt}$$

$$2 = \frac{1}{2} \frac{dx}{dt} (10) + \frac{1}{2} (20) (1)$$

$$2 = 5 \frac{dx}{dt} + 10$$

$$2 - 10 = 5 \frac{dx}{dt}$$

$$\div 5 \quad 5 \frac{dx}{dt} = -8$$

$$\frac{dx}{dt} = -\frac{8}{5} \text{ cm/min}$$

$$A = \frac{1}{2}xy$$

$$100 = \frac{1}{2}x(10)$$

$$100 = 5x$$

$$\div 5$$

$$x = 20 \text{ cm}$$

Air is being pumped into spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$.

How fast is the radius of the balloon is increasing when the radius of radius is 25 cm ?

Apr 2019

$$\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$$

$$\frac{dr}{dt} ?? \longrightarrow r = 25 \text{ cm}$$

$$V = \frac{4}{3} \pi r^3$$

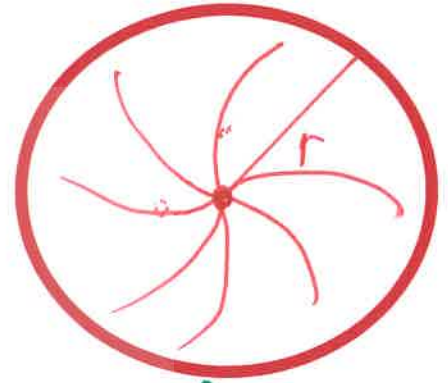
Diff. w.r.t. t

$$\frac{dV}{dt} = \frac{4}{3} \pi (3r^2) \frac{dr}{dt}$$

$$100 = 4\pi (25)^2 \frac{dr}{dt}$$

$$\div 4\pi (25)^2$$

$$\begin{aligned} \frac{dr}{dt} &= \frac{100}{4\pi (25)^2} = \frac{25}{\pi (25)^2} \\ &= \frac{1}{\pi (25)} \text{ cm/s} \end{aligned}$$



$$V = \frac{4}{3} \pi r^3 \quad S = 4\pi r^2$$

The side x , y and z of closed rectangular box are changing with respect to time in such that

$$\frac{dx}{dt} = 1 \text{ m/sec}, \quad \frac{dy}{dt} = -2 \text{ m/sec} \quad \text{and} \quad \frac{dz}{dt} = 2 \text{ m/sec}$$

At what rate is the surface area of box changing when $x = 4 \text{ m}$, $y = 3 \text{ m}$ and $z = 2 \text{ m}$

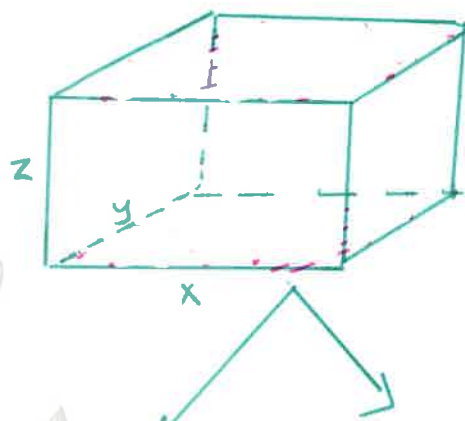
$$\frac{dx}{dt} = 1 \text{ m/sec}$$

$$\frac{dy}{dt} = -2 \text{ m/sec}$$

$$\frac{dz}{dt} = 2 \text{ m/sec}$$

$$\frac{ds}{dt} ??$$

$$\begin{aligned} x &= 4 \text{ m} \\ y &= 3 \text{ m} \\ z &= 2 \text{ m} \end{aligned}$$



$$V = x \cdot y \cdot z$$

$$S = 2xy + 2xz + 2yz$$

$$S = 2xy + 2xz + 2yz$$

Diff. w.r.t. t

$$\begin{aligned} \frac{ds}{dt} &= 2(1) \frac{dx}{dt} y + 2x(1) \frac{dy}{dt} + 2(1) \frac{dx}{dt} z + 2x(1) \frac{dz}{dt} \\ &\quad + 2(1) \frac{dy}{dt} z + 2y(1) \frac{dz}{dt} \end{aligned}$$

$$\begin{aligned} &= 2(1)(3) + 2(4)(-2) + 2(1)(2) + 2(4)(2) \\ &\quad + 2(-2)(2) + 2(3)(2) \end{aligned}$$

$$= 6 - 16 + 4 + 16 - 8 + 12 = 14 \text{ m}^2/\text{sec}.$$

The length L of a rectangle is decreasing at rate of 2cm/sec while the width W is increasing at rate of 1cm/sec .

Find the rate of change of the area A of the rectangle when $L = 12\text{cm}$ and $W = 5\text{cm}$

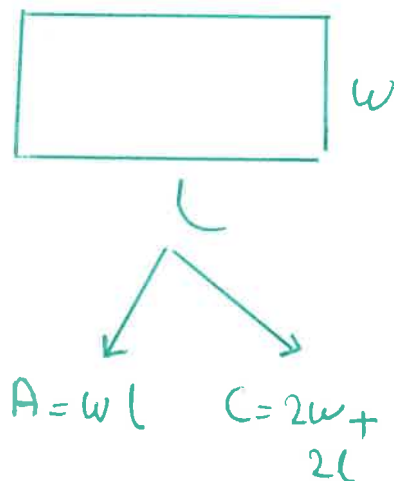
$$\frac{dL}{dt} = -2\text{cm/sec.}$$

$$\frac{dW}{dt} = 1\text{cm/sec}$$

$$\frac{dA}{dt} ??$$

$$L = 12\text{cm}$$

$$W = 5\text{cm}$$



$$A = w \cdot L$$

Diff. w.r.t. t

$$\frac{dA}{dt} = (1) \frac{dw}{dt} L + w (1) \frac{dL}{dt}$$

$$= (1)(12) + (5)(-2)$$

$$= 12 - 10$$

$$= 2\text{ cm}^2/\text{Sec.}$$

As a right circular cylinder metal rod is being heated , its height is increasing at rate of $0.002\text{cm}/\text{min}$ and its radius is increasing at a rate $0.001\text{cm}/\text{min}$. At what rate is the volume changing when the rod has height 20 cm and radius 4cm .

$$\frac{dh}{dt} = 0.002 \text{ cm/min}$$

$$\frac{dr}{dt} = 0.001 \text{ cm/min}$$

$$\frac{dV}{dt} \quad ?? \rightarrow \begin{matrix} h = 20 \text{ cm} \\ r = 4 \text{ cm} \end{matrix}$$

$$V = \pi r^2 h$$

Diff. w.r.t. t

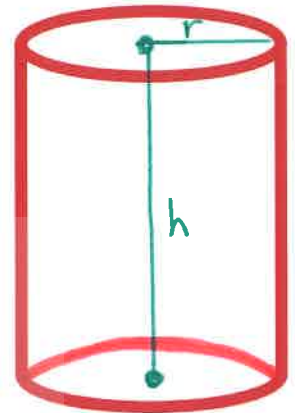
$$(1) \frac{dV}{dt} = \pi (2r) \frac{dr}{dt} h + \pi r^2 (1) \frac{dh}{dt}$$

$$= \pi (8) (0.001) (20) + \pi (16) (0.002)$$

$$= \pi (160)(0.001) + \pi (16)(0.002)$$

$$= \pi (0.160) + (0.032)$$

$$= \pi (0.192) \text{ cm}^3/\text{min}$$



$$V = \pi r^2 h \quad S = 2\pi r h$$

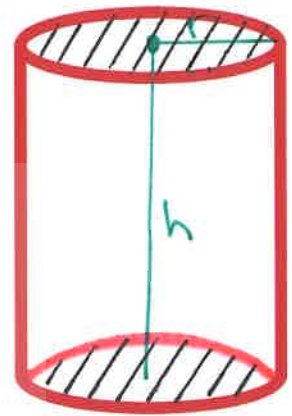
The radius of a closed right circular cylinder is decreasing at a rate of 1cm/sec and the height is increasing at a rate of 4cm/sec .

Find the rate at which the total surface area of the cylinder is changing when the radius is 6 cm and the height is 10cm .

$$\frac{dr}{dt} = \overset{\text{decreasing}}{-1} \text{ cm/sec}$$

$$\frac{dh}{dt} = 4 \text{ cm/sec}$$

$$\frac{dS}{dt} ?? \rightarrow \begin{matrix} r = 6 \text{ cm} \\ h = 10 \text{ cm} \end{matrix}$$



$$V = \pi r^2 h \quad S = 2\pi rh + 2\pi r^2$$

$$S = 2\pi rh + 2\pi r^2$$

Diff. w.r.t. t

$$\frac{dS}{dt} = 2\pi \frac{dr}{dt} h + 2\pi r \frac{dh}{dt} + 4\pi r \frac{dr}{dt}$$

$$= 2\pi (-1)(10) + 2\pi (6)(4) + 4\pi (6)(-1)$$

$$= -20\pi + 48\pi - 24\pi$$

$$= 4\pi \text{ cm}^2/\text{sec}.$$

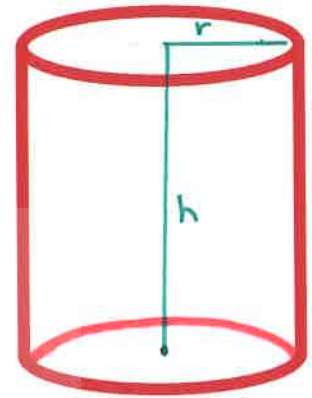
A cylindrical tank of diameter 6 m is being filled with water at a rate of $2 \text{ m}^3/\text{hr}$, how fast is the water level rising ?

Dec 2016

$$d = 6 \text{ m} \rightsquigarrow r = 3 \text{ m}$$

$$\frac{dV}{dt} = 2 \text{ m}^3/\text{hr}$$

$$\frac{dh}{dt} ??$$



$$V = \pi r^2 h \quad S = 2\pi r h$$

$$V = \pi r^2 h$$

$$V = \pi (3)^2 h$$

$$V = 9\pi h$$

Diff. w.r.t. t

$$\frac{dV}{dt} = 9\pi \frac{dh}{dt}$$

$$2 = 9\pi \frac{dh}{dt}$$

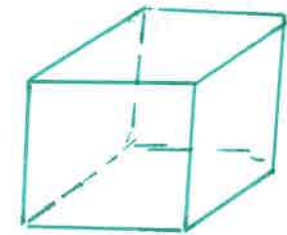
$$\frac{dh}{dt} = \frac{2}{9\pi} \text{ m/hr}$$

The volume of the cube is increasing at a rate of $30 \text{ cm}^3/\text{sec}$

How fast is the total surface area of the cube increasing when the length of each edge of the cube is 10 cm long.

$$\frac{dV}{dt} = 30 \text{ cm}^3/\text{sec}$$

$$\frac{dS}{dt} ?? \xrightarrow{\text{when}} x = 10 \text{ cm}$$



$$V = x^3$$

$$S = 6x^2$$

$$S = 6x^2$$

Diff. w.r.t. t

$$\frac{dS}{dt} = 12x \frac{dx}{dt}$$

?

$$= 12(10) \frac{1}{10}$$

$$= 12 \text{ cm}^2/\text{sec}$$

$$V = x^3$$

Diff. w.r.t. t

$$\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$$

$$30 = 3(10)^2 \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{30}{3(10)^2}$$

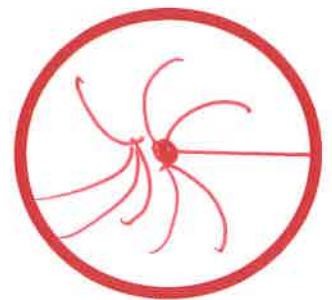
$$= \frac{30}{3(100)}$$

$$= \frac{10}{100} = \frac{1}{10}$$

A snow ball is melting at a rate of $0.03 \text{ ft}^3/\text{hr}$. At what rate is the surface area changing when the volume of the ball is $36\pi \text{ ft}^3$?

$$\frac{dV}{dt} = \overset{\text{melting}}{-} 0.03 \text{ ft}^3/\text{hr}$$

$$\frac{dS}{dt} ?? \longrightarrow V = 36\pi \text{ ft}^3$$



$$V = \frac{4}{3}\pi r^3 \quad S = 4\pi r^2$$

Diff. w.r.t. t

$$\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

$$= \cancel{8\pi}^4 (3) \frac{-0.03}{\cancel{36\pi}^{18}}$$

$$= \cancel{4}^2 (3) \frac{-0.03}{\cancel{18}^9}$$

$$= \cancel{2}^1 (3) \frac{-0.03}{\cancel{9}^3}$$

$$= 2 \frac{-0.03}{3}$$

$$= 2(-0.01)$$

$$= -0.02 \text{ ft}^2/\text{hr}$$

$$V = \frac{4}{3}\pi r^3$$

$$36\pi = \frac{4\pi}{3} r^3$$

$$\times \frac{3}{4\pi}$$

$$\frac{3}{4\pi} 36\pi = \frac{3}{4\pi} \frac{4\pi}{3} r^3$$

$$3(9) = r^3$$

$$r^3 = 27 \longrightarrow r = 3 \text{ ft}$$

$$V = \frac{4}{3}\pi r^3$$

Diff. w.r.t. t

$$\frac{dV}{dt} = \frac{4}{3}\pi 3r^2 \frac{dr}{dt}$$

$$-0.03 = 4\pi (9) \frac{dr}{dt}$$

$$-0.03 = 36\pi \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{-0.03}{36\pi} \text{ ft/hr}$$

A point P moves on the curve $y = x + 5$ such that

$\frac{dx}{dt} = 3 \text{ units/sec}$. Find the rate of the change of the distance

between P and the point $Q(2, 0)$ when P is at $(-5, 0)$.

$$y = x + 5$$

$$Q(2, 0)$$

$$\frac{dx}{dt} = 3 \text{ units/sec}$$

$$\frac{dL}{dt} ?? \longrightarrow P(-5, 0)$$

$$L^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$L^2 = (x - 2)^2 + (y - 0)^2$$

$$L^2 = (x - 2)^2 + y^2$$

$$L^2 = (x - 2)^2 + (x + 5)^2$$

Diff. w.r.t. t

$$2L \frac{dL}{dt} = 2(x - 2) \frac{dx}{dt} + 2(x + 5) \frac{dx}{dt}$$

$$2(7) \frac{dL}{dt} = 2(-5 - 2)(3) + 0$$

$$14 \frac{dL}{dt} = 2(-7)(3)$$

$\div 14$

$$\frac{dL}{dt} = \frac{2(-7)(3)}{14}$$

$$= -3 \text{ unit/sec.}$$

$$L^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$L^2 = (2 - (-5))^2 + (0 - 0)^2$$

$$L^2 = (7)^2$$

$$L = 7 \text{ unit}$$

A point $P(x, y)$ moves on the line $y = 2x - 1$, so that the

$$\frac{dx}{dt} = 5 \text{ cm/sec}$$

Find the rate at which the distance between P and $Q(-1, 1)$ is changing when $P(0, 1)$

$$y = 2x - 1$$

$$Q(-1, 1)$$

$$\frac{dx}{dt} = 5 \text{ cm/sec}$$

$$\frac{dL}{dt} ?? \text{ when } \rightarrow P(0, 1)$$

$$L^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$L^2 = (x - (-1))^2 + (y - 1)^2$$

$$L^2 = (x + 1)^2 + (y - 1)^2$$

$$L^2 = (x + 1)^2 + (2x - 1 - 1)^2$$

$$L^2 = (x + 1)^2 + (2x - 2)^2$$

Diff. w.r.t. t

$$2L \frac{dL}{dt} = 2(x + 1) \frac{dx}{dt} + 2(2x - 2)(2) \frac{dx}{dt}$$

$$2(1) \frac{dL}{dt} = 2(0 + 1)(5) + 2(0 - 2)(2)(5)$$

$$2 \frac{dL}{dt} = 10 - 40$$

$$2 \frac{dL}{dt} = -30 \rightarrow \frac{dL}{dt} = -15 \text{ cm/sec}$$

$$L^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$L^2 = (-1 - 0)^2 + (1 - 1)^2$$

$$L^2 = (-1)^2$$

$$L^2 = 1$$

$$L = 1$$

A point $P(x, y)$ is moving on the curve $y = x^2 - 3$, such that the x - *coordinate* is increasing at rate of 7cm/sec

what is the rate of change of y - *coordinate* when the point P reaches $(3, 6)$

$$y = x^2 - 3$$

$$\frac{dx}{dt} = 7 \text{ cm/sec}$$

$$\frac{dy}{dt} ?? \text{ when } \rightarrow P(3, 6)$$

$$y = x^2 - 3$$

Diff. w.r.t. t

$$\frac{dy}{dt} = 2x \frac{dx}{dt}$$

$$\frac{dy}{dt} = 2(3)(7)$$

$$\frac{dy}{dt} = 42 \text{ cm/sec}$$

Two students start walking from the same point. One walks south at a rate of 4 m/s and the other walks west at a rate of 3 m/s .

At what rate is the distance between the two students increasing 2 seconds later.

$$\frac{dy}{dt} = 4 \text{ m/s}$$

$$\frac{dx}{dt} = 3 \text{ m/s}$$

$$\frac{dz}{dt} ?? \rightarrow t = 2 \text{ sec.}$$

$$z^2 = x^2 + y^2$$

Diff. w.r.t. t

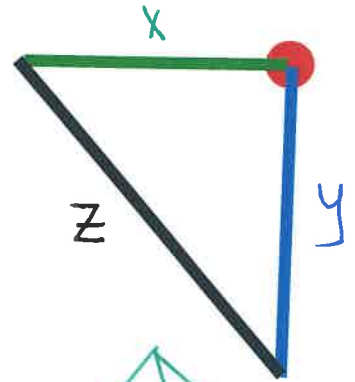
$$2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$2(10) \frac{dz}{dt} = 2(6)(3) + 2(8)(4)$$

$$20 \frac{dz}{dt} = 36 + 64$$

$$20 \frac{dz}{dt} = 100$$

$$\div 20 \quad \frac{dz}{dt} = \frac{100}{20} = 5 \text{ m/s}$$



$$A = \frac{1}{2}xy \quad C = x + y + z \quad z^2 = x^2 + y^2$$

الزمن . السرعة = المسافة

$$x = \frac{dx}{dt} \cdot t$$

$$= 3(2) = 6\text{ m}$$

$$y = \frac{dy}{dt} \cdot t$$

$$= 4(2) = 8\text{ m}$$

$$z^2 = x^2 + y^2$$

$$z^2 = 6^2 + 8^2$$

$$z^2 = 36 + 64 = 100$$

$$z = 10$$

Two people start running from the same point P . One runs east at 6 km/h and the other runs north at 8 km/h .

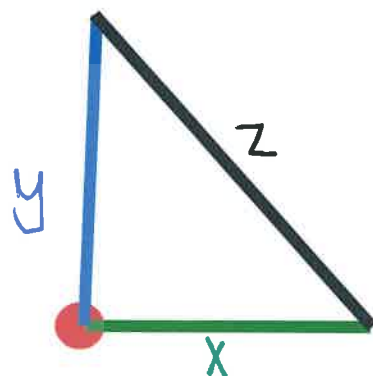
How fast is the distance between the two people changing after 0.5 hour ?

May 2017

$$\frac{dx}{dt} = 6 \text{ km/h}$$

$$\frac{dy}{dt} = 8 \text{ km/h}$$

$$\frac{dz}{dt} ?? \rightarrow t = 0.5 \text{ hr}$$



$$z^2 = x^2 + y^2$$

Diff. w.r.t. t

$$2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$2(5) \frac{dz}{dt} = 2(3)(6) + 2(4)(8)$$

$$10 \frac{dz}{dt} = 36 + 64$$

$$10 \frac{dz}{dt} = 100$$

$\div 10$

$$\frac{dz}{dt} = 10 \text{ km/h}$$

$$x = \frac{dx}{dt} \cdot t$$

$$= 6(0.5)$$

$$x = 3 \text{ km}$$

$$y = \frac{dy}{dt} \cdot t$$

$$= 8(0.5)$$

$$y = 4 \text{ km}$$

$$z^2 = x^2 + y^2$$

$$z^2 = 9 + 16 = 25$$

$$z = 5 \text{ km}$$

Two boats (A and B) are heading to the same intersection point (C).
 Boat A is sailing west at speed 50 *mile/hour*, while boat B is sailing north at speed 60 *mile/hour*. At what rate are these boats approaching each other when boat A is 0.3 *mile* and boat B is 0.4 *mile* from the intersection point C

$$\frac{dx}{dt} = -50 \text{ mile/hr}$$

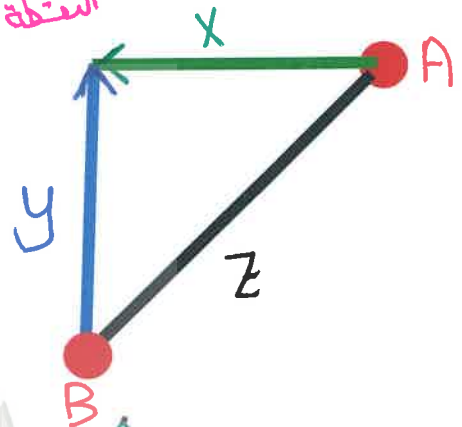
$$\frac{dy}{dt} = -60 \text{ mile/hr}$$

$$\frac{dz}{dt} ??$$

$$x = 0.3 \text{ mile}$$

$$y = 0.4 \text{ mile}$$

نام يقعون لنفس النقطة



$$z^2 = x^2 + y^2$$

Diff. w.r.t. t

$$2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$2(0.5) \frac{dz}{dt} = 2(0.3)(-50) + 2(0.4)(-60)$$

$$1.0 \frac{dz}{dt} = -100(0.3) - 120(0.4)$$

$$\frac{dz}{dt} = -30.0 - 48.0$$

$$\frac{dz}{dt} = -78 \text{ mile/hr}$$

$$A = \frac{1}{2}xy$$

$$C = x + y + z$$

$$z^2 = x^2 + y^2$$

$$z^2 = x^2 + y^2$$

$$z^2 = (0.3)^2 + (0.4)^2$$

$$z^2 = 0.09 + 0.16$$

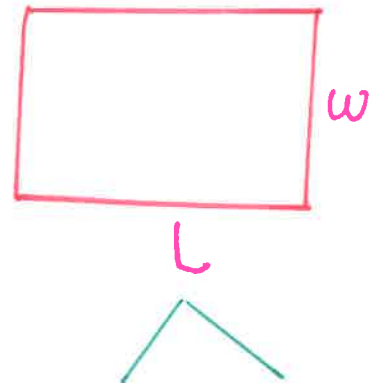
$$z^2 = 0.25$$

$$z = 0.5 \text{ mile}$$

The width of rectangle is half its length . at what rate is the area of the rectangle increasing when the width is 10cm and it is increasing at a rate of 0.5cm/sec

$$w = \frac{1}{2} L \longleftrightarrow L = 2w$$

$$\frac{dA}{dt} ?? \longrightarrow w = 10 \text{ cm}$$
$$\frac{dw}{dt} = 0.5 \text{ cm/sec}$$



$$A = wL \quad C = 2w + 2L$$

$$A = w \cdot L$$

$$A = w (2w)$$

$$A = 2w^2$$

Diff. w.r.t .t

$$\frac{dA}{dt} = 4w \frac{dw}{dt}$$

$$= 4(10)(0.5)$$

$$= 4(10) \frac{1}{2}$$

$$= 4(5) = 20 \text{ cm}^2/\text{Sec}$$

A cone of metal whose altitude is twice its base radius is heated so that its base radius is increasing at rate of 0.01 cm/min . Find the rate at which its volume is changing when the base radius is 10 cm .

$$h = 2r \quad \rightsquigarrow \quad r = \frac{h}{2}$$

$$\frac{dr}{dt} = 0.01 \text{ cm/min}$$

$$\frac{dV}{dt} \quad ?? \quad \longrightarrow \quad r = 10 \text{ cm}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi r^2 (2r)$$

$$V = \frac{2}{3} \pi r^3$$

Diff. w.r.t. t

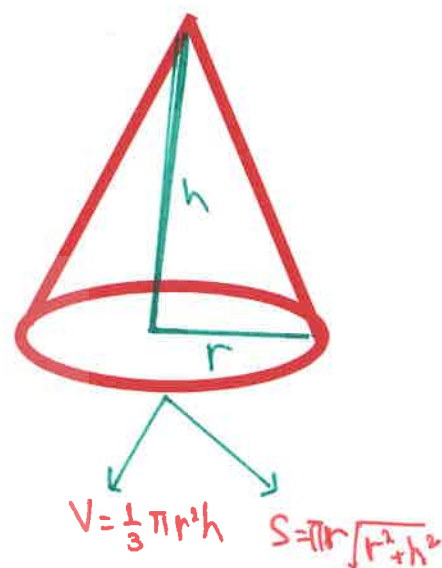
$$\frac{dV}{dt} = \frac{2}{3} \pi \cancel{3} r^2 \frac{dr}{dt}$$

$$= 2\pi (10)^2 (0.01)$$

$$= 2\pi (100) (0.01)$$

$$= 2\pi \quad 1.00$$

$$= 2\pi \text{ cm}^3/\text{min}$$



A cone of ice cream whose altitude is three its base radius is melting without losing shape at a rate of $0.3 \text{ cm}^3/\text{min}$.

Find the rate at which its altitude is changing when its radius is 2 cm .

$$h = 3r \rightsquigarrow r = \frac{h}{3}$$

$$\frac{dV}{dt} = \overset{\text{melting}}{-0.3} \text{ cm}^3/\text{min}$$

$$\frac{dh}{dt} ?? \rightarrow r = 2 \text{ cm}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi \left(\frac{h}{3}\right)^2 h$$

$$= \frac{1}{3} \pi \frac{h^2}{9} h$$

$$V = \frac{1}{27} \pi h^3$$

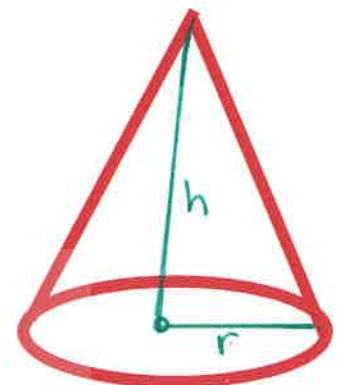
Diff. w.r.t. t

$$\frac{dV}{dt} = \frac{1}{27} \pi \cancel{3} h^2 \frac{dh}{dt}$$

$$-0.3 = \frac{1}{9} \pi (6)^2 \frac{dh}{dt}$$

$$\div 4\pi \quad -0.3 = \frac{1}{9} \pi \cdot 36 \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{-0.3}{4\pi} \text{ cm/min}$$



$$V = \frac{1}{3} \pi r^2 h \quad S = \pi r \sqrt{r^2 + h^2}$$

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$$\text{at } r = 2$$

$$h = 3(2)$$

$$h = 6 \text{ cm}$$



Tricky question !!



A water tank has the shape of an inverted circular cone with base radius 2 m and the height 4 m . If water is being pumped into the tank at a rate of $2\text{ m}^3/\text{min}$. Find the rate at which the water level is rising when the water is 3 m deep.

$$r = 2\text{ m}$$

$$h = 4\text{ m}$$

$$\frac{r}{h} = \frac{2}{4} = \frac{1}{2}$$

$$\frac{r}{h} = \frac{1}{2}$$

$$2r = h \rightarrow r = \frac{h}{2}$$

$$\frac{dV}{dt} = 2\text{ m}^3/\text{min}$$

$$\frac{dh}{dt} ?? \rightarrow h = 3\text{ m}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi \left(\frac{h}{2}\right)^2 h$$

$$V = \frac{1}{3} \pi \frac{h^2}{4} h$$

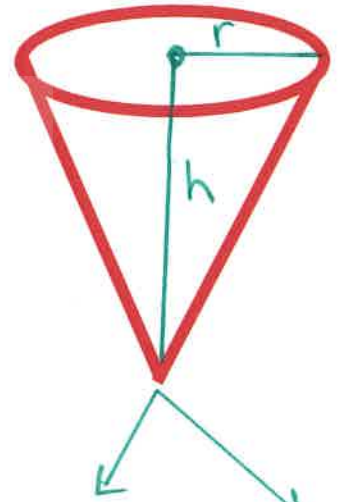
$$V = \frac{1}{12} \pi h^3$$

Diff. w.r.t. t

$$\frac{dV}{dt} = \frac{1}{12} \pi 3h^2 \frac{dh}{dt}$$

$$2 = \frac{1}{4} \pi (3)^2 \frac{dh}{dt}$$

$$2 = \frac{9\pi}{4} \frac{dh}{dt} \rightarrow \frac{dh}{dt} = 2 \frac{4}{9\pi} = \frac{8}{9\pi} \text{ m/min}$$



$$V = \frac{1}{3} \pi r^2 h \quad S = \pi r \sqrt{r^2 + h^2}$$

A ladder 13 ft long is leaning against the side of the building. If the bottom of the ladder is pulled away from the building at a rate of 0.1 ft/sec. How fast is the angle formed by the ladder and the ground changing at the instant when the top of the ladder is 12 ft above the ground.

$$Z = 13 \text{ ft}$$

$$\frac{dx}{dt} = 0.1 \text{ ft/sec}$$

$$\frac{d\theta}{dt} ?? \rightarrow y = 12 \text{ ft}$$

$$\cos \theta = \frac{x}{z}$$

$$\cos \theta = \frac{x}{13}$$

Diff. w.r.t. t

$$-\sin \theta \frac{d\theta}{dt} = \frac{1}{13} \frac{dx}{dt}$$

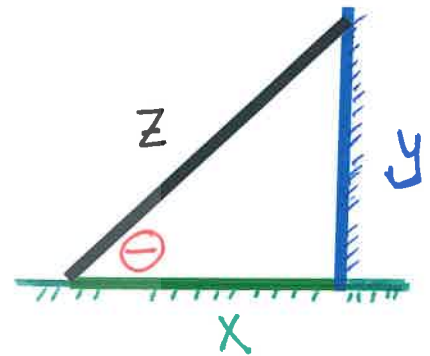
$$\sin \theta = \frac{y}{z}$$

$$\sin \theta = \frac{12}{13}$$

$$-\frac{12}{13} \frac{d\theta}{dt} = \frac{1}{13} (0.1)$$

$$-\frac{12}{13} \frac{d\theta}{dt} = \frac{1}{13} \cdot \frac{1}{10}$$

$$* \frac{-13}{12} \frac{d\theta}{dt} = \frac{-13}{12} \cdot \frac{1}{13} \cdot \frac{1}{10} = \frac{-1}{120}$$



A rectangle has a fixed area of 24 cm^2 . The length is decreasing at a rate of 2 cm/min . At what rate the width is increasing when the length is 8 cm .

Nov 2018

$$A = 24 \text{ cm}^2$$

$$\frac{dL}{dt} = -2 \text{ cm/min}$$

$$\frac{dw}{dt} ?? \rightarrow L = 8 \text{ cm}$$

$$A = L \cdot w$$

$$24 = L \cdot w$$

Diff. w.r.t. t

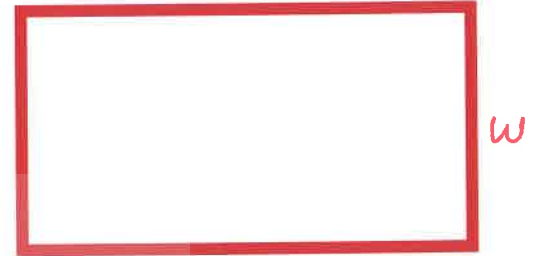
$$0 = (1) \frac{dL}{dt} w + L (1) \frac{dw}{dt}$$

$$0 = -2(3) + 8 \frac{dw}{dt}$$

$$6 = 8 \frac{dw}{dt}$$

$\div 8$

$$\frac{dw}{dt} = \frac{6}{8} = \frac{3}{4} \text{ cm/min}$$



$$A = L \cdot w \quad C = 2L + 2w$$

$$A = L \cdot w$$

$$24 = 8 \cdot w$$

$$w = \frac{24}{8}$$

$$w = 3 \text{ cm}$$